

AMERICAN UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER AND COMMUNICATIONS ENGINEERING

CIRCUIT ANALYSIS 1 LAB

CCE 201L

SUMMER HOUSE

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ABSTRACT

This project presents the design, control logic, and simulation of the "Summer House," an automated smart home system designed to enhance convenience and energy efficiency through discrete analog and mixed-signal electronics. The system incorporates three primary automated sub-functions: a temperature-controlled cooling fan, an automated swimming pool filling pump, and a water-level status indicator. Using an LM358 operational amplifier configured as a comparator alongside temperature and water sensing circuitry, the design eliminates the need for complex digital microcontrollers for basic automation. Simulation results created in Proteus demonstrate that the fan activates automatically when room temperature exceeds 30°C, the pool pump operates when triggered under empty-pool conditions, and LED indicator branches clearly display system fluid levels. The project establishes a functional foundation for simple electronic automation and outlines operational pathways for future microcontroller-based expansion.

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I. Objectives

The primary objective of this project is to develop an automated control system for a "Summer House" model using fundamental electronic components. The specific sub-objectives are:

Temperature-Controlled Ventilation: To implement an automated fan control subsystem that triggers active cooling when the ambient temperature exceeds a pre-set threshold of 30°C.

Automated Pool Refilling: To design a logic-driven pool pump actuation circuit that fills the pool automatically upon switch activation, provided the pool is detected as empty.

Visual Water Level Indication: To construct a discrete transistor-based water-level indicator that provides visual feedback on tank/pool fill status.

II. BACKGROUND THEORY

A. Temperature Sensing via Thermistors

Thermistors are temperature-sensitive resistors whose resistance varies significantly with temperature change. In a Negative Temperature Coefficient (NTC) thermistor, resistance decreases as temperature increases. When placed in a voltage divider network, variations in ambient temperature alter the output voltage at the junction. This changing voltage serves as an analog input to a threshold-detection circuit.

B. Operational Amplifier as a Voltage Comparator

An operational amplifier (Op-Amp) operating without negative feedback acts as a high-gain voltage comparator. The device compares the non-inverting input voltage (V_+) with the inverting input voltage (V_-):The LM358 is a dual low-power operational amplifier IC capable of operating from a single power supply. It is well-suited for single-supply comparator applications in low-voltage automation systems.

C. Transistor Switching & Inductive Load Driving

Bipolar Junction Transistors (BJTs), such as NPN transistors, function as electronic switches when driven into saturation. A small current at the base drives the collector-emitter path into conduction, enabling low-power operational amplifier outputs to control higher-current loads like DC fans and pumps.

When switching inductive loads (e.g., DC motors, pumps, and fans), a flyback diode connected in parallel across the load protects the driving transistor from damaging inductive voltage spikes generated when the current is rapidly switched off.

III. REQUIRED COMPONENTS

The complete system implementation relies on the components specified in Table 1 below:

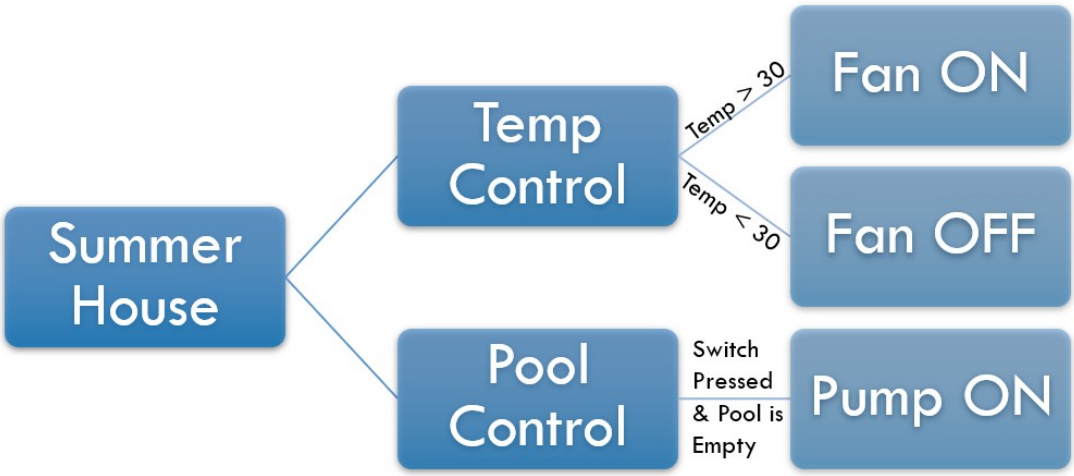
Table 1. Required Components and System Roles**

Component Name	Part / Type	Role / Function in Project
Operational Amplifier	LM358	Dual op-amp IC used as voltage comparators for fan and pump control.
Cooling Fan	5V DC Brushless Fan	Provides active cooling when triggered by high ambient temperature.
Water Pump	12V Brushless DC Pump	Fills the pool automatically when activated under empty conditions.
Temperature Sensor	NTC Thermistor (103 / 10k)	Detects ambient room temperature within a resistive divider network.
BJT Transistors	NPN Transistors (2N3904)	Act as electronic switches for motor loads and LED indicator branches.
Resistors	Fixed & Potentiometers	Set voltage reference dividers, base current limiters, and LED drop resistors.

IV. SYSTEM ARCHITECTURE

The overall architecture of the Summer House automation system is divided into three distinct operational domains governed by analog decision rules:

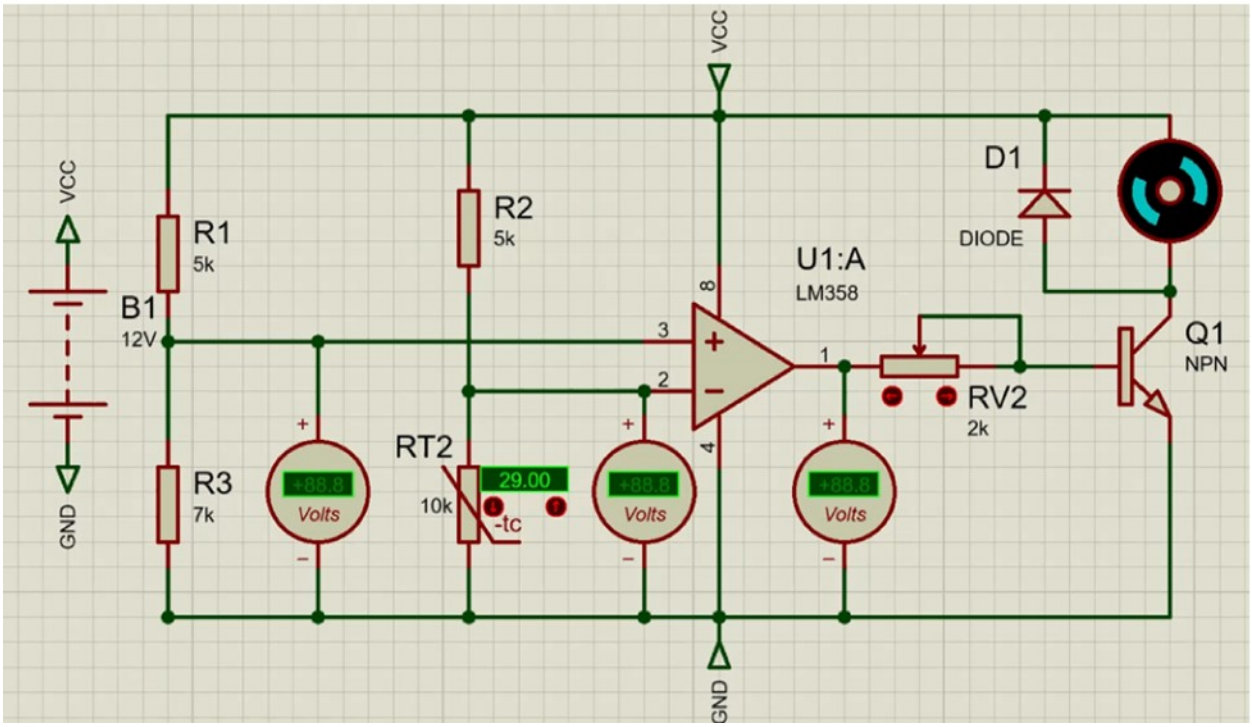
Figure 1. High-Level System Architecture and Operational Rules



1. Temperature Control Subsystem: Continuously monitors room temperature. If T > 30 , the DC fan turns ON. If $T < 30, the fan remains OFF.
2. Pool Pump Subsystem: Controls pool filling. The 12V pump activates only when a manual fill switch is pressed AND the system senses that the pool is empty.
3. Water Level Indicator Subsystem: Displays fluid level states through discrete LED outputs.

V. SCHEMATIC REPRESENTATION & CIRCUIT ANALYSIS

A. Temperature-Based Fan Control Circuit



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Figure 2. Temperature-Based Fan Control Schematic and Simulation (T = 29, Fan OFF)

Reference Voltage ($V_+ at Pin 3):Fixed by a voltage divider formed by R_1 = 5 and R_3 = 7.

Sensing Voltage ($V_- at Pin 2):Formed by resistor R_2 = 5and thermistor RT_2 = 10k (NTC). As temperature rises, RT_2 resistance decreases, lowering the voltage at Pin 2.

Switching Stage: When temperature exceeds 30℃, V_+ > V_-, forcing Pin 1 HIGH. Current flows through variable resistor RV_2 into the base of transistor Q_1, switching on the DC fan. Flyback diode D_1suppresses inductive voltage spikes.

B. Automatic Pool Pump Control Circuit

The pool pump control system relies on reference voltage levels established by fixed resistor dividers that represent water level conditions.

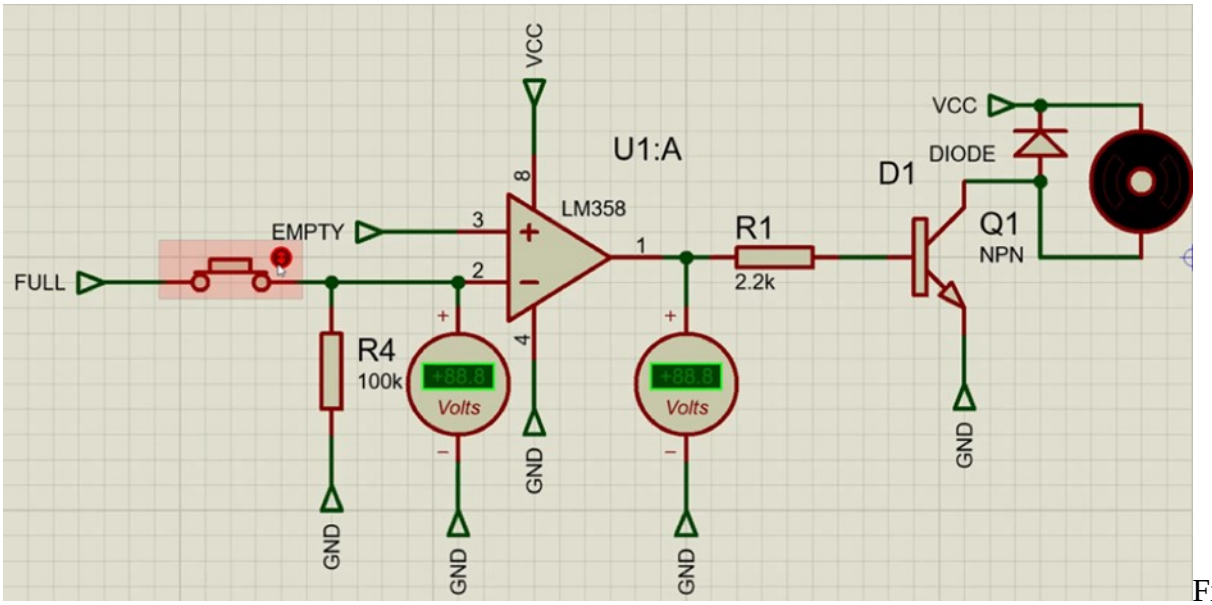


Figure 3. Pool Level Reference Voltage Divider Circuits (FULL and EMPTY nodes)*

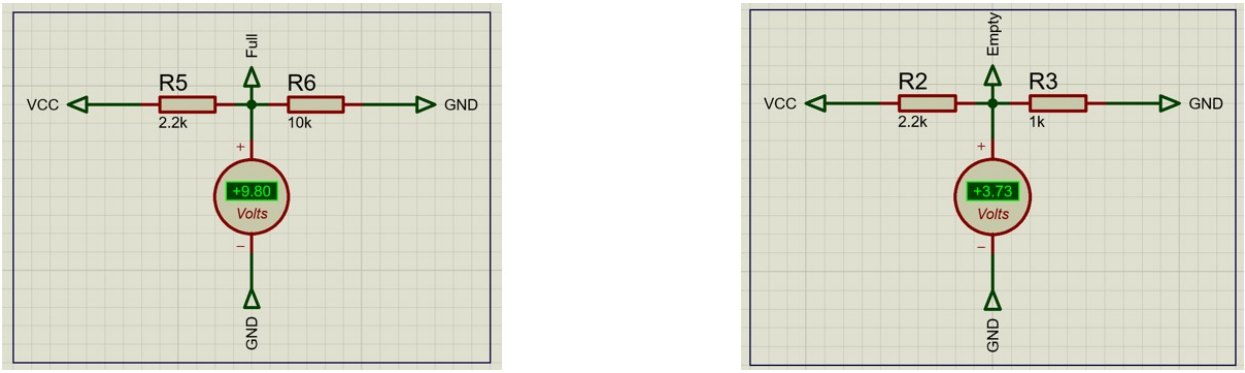


Figure 4. Automatic Pool Pump Control Schematic and Simulation

Voltage Reference Dividers:The `FULL` node provides a high voltage signal (+9.80V) via divider R_5/R_6, while the `EMPTY` node provides a low voltage signal (+3.73V) via divider R_2/R_3.

Comparator Logic (LM358 U1:A):The `EMPTY` voltage (+3.73V) feeds the non-inverting input (Pin 3). The inverting input (Pin 2) is tied to the `FULL` signal through a push-button switch with pull-down resistor $R_4 = 100Pump Actuation:When the pool is empty and the switch open/pressed under empty conditions, Pin 3 remains higher than Pin 2, driving Output Pi1 HIGH. Base current flows through R_1 = 2.2into transistor Q_1, driving the 12V pump motor.

C. Water Level Indicator Circuit

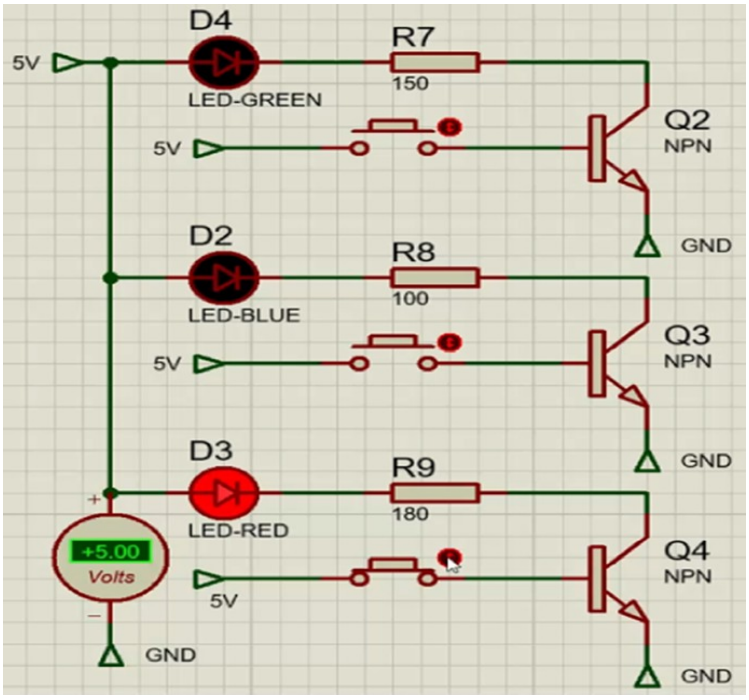


Figure 5. Transistor-Driven Water Level Indicator Circuit Schematic

The indicator circuit consists of three independent NPN transistor switching branches operating from a +5V supply:

High Level Branch: Drives a Green LED (D_4) through current-limiting resistor R_7 = 150 when transistor Q_2 base is energized.

Medium Level Branch: Drives a Blue LED (D_2) through R_8 = 100 transistor Q_3.

Low/Empty Level Branch:** Drives a Red LED (D_3) through R_9 = 180via transistor Q_4.

VI. PROCEDURE & SIMULATION FRAMEWORK

A. Fan Control Simulation Procedure

1. Construct the fan control circuit in Proteus using an LM358 operational amplifier, a 10k NTC thermistor model, fixed biasing resistors (R_1, R_2, R_3), an NPN transistor, a 2k potentiometer, a flyback diode, and a DC motor.

2. Set the fixed reference voltage at Pin 3 using the R_1/R_3 voltage divider.

3. Adjust the simulated thermistor temperature below 30°C ($T = 29^\\circ\text{C}$) and verify that the comparator output remains LOW, keeping the fan turned off.

4. Increase the temperature above 30°C and observe output switching, current flow through the transistor base, and active motor rotation.

B. Pool Pump Simulation Procedure

1. Wire the static voltage divider networks in Proteus to establish reference points representing `FULL` (+9.80V) and `EMPTY` (+3.73V) states.

2. Connect these reference nodes to the inputs of the LM358 comparator along with a push-button switch and a $100\text{ k}\Omega$ pull-down resistor.

3. Simulate activation of the pump control switch under empty pool conditions and confirm that the output drives the NPN transistor switch into saturation to power the 12V DC pump.

C. Water Level Indicator Simulation Procedure

1. Build three transistor switching branches in Proteus using 2N3904 NPN transistors, LEDs (Green, Blue, Red), current-limiting resistors ($150\,\Omega, 100\,\Omega, 180\,\Omega$), and input signal switches connected to a +5V rail.

2. Toggle each switch independently to verify that base current triggers the corresponding collector-emitter path, turning on the respective LED indicator.

VII. RESULTS & DISCUSSION

The Proteus software simulations verified all planned control logic functions without requiring digital microcontrollers:

1. Fan Control Performance:At T = 29.00the voltage at Pin 2 remains higher than Pin 3, holding the output at 0V and keeping transistor Q_1 off (DC fan idle). Raising the temperature above 30°C drops thermistor resistance RT_2, causing Pin 3 voltage to exceed Pin 2. The output switches HIGH, driving Q_1 into saturation and turning on the fan.

2. **Pool Pump Performance:** The voltage dividers established accurate logic reference levels of +9.80V (`FULL`) and +3.73V (`EMPTY`). When the pool was empty and the activation switch was closed, the comparator successfully output a HIGH signal, activating the 12V DC pump motor.

3. **Water Level Indicator Performance:** All three indicator branches operated correctly. Closing each sensing switch applied base current to its respective NPN transistor, lighting the corresponding status LED (Red = Empty, Blue = Medium, Green = Full).

VIII. CONCLUSION & FUTURE WORK

The "Summer House" project demonstrates that simple electronic circuits can effectively automate core residential sub-systems. Using operational amplifiers as comparators and transistors as electronic switches, the design successfully delivers automated temperature-controlled fan ventilation, pool refilling, and water level indication without digital microcontrollers.

For future development, the system could be expanded by integrating a microcontroller (such as an Arduino or PIC microcontroller). Adding a programmable controller would allow for fine-grained PID temperature regulation, multi-sensor fluid monitoring, variable pump timing, and smart user interfaces.

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[3] "LM358 Low-Power Dual Operational Amplifiers Datasheet," Literature No. SLOS068W, Revised 2021.